

Detailed Point-by-Point Response to Reviewers

Paper ID 7809: Frequency-Guided Semantic Steganography for High-Fidelity Self-Contained Style-Transfer Image Hiding

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We sincerely thank the Program Chairs and the Reviewers for their constructive, thorough, and valuable feedback. In accordance with the reviewers' recommendations, we have revised the manuscript. For transparency and ease of verification, all revised or newly added text in the updated manuscript is printed in **blue font**.

Below is our detailed, point-by-point response to all comments and questions.

Response to Reviewer #1

Reviewer Comment

Point 1: Component validation and systematic ablation. The proposed framework contains several interesting components, but the contribution of each component is not sufficiently validated. On Pages 4–8, FGSS combines semantic tokenisation, texture-guided masking, frequency-domain regularisation, a nonlinear capacity-ceiling penalty, oracle distillation, serial consistency, and a multi-stage training schedule... A systematic ablation on the main evaluation set would considerably strengthen the paper. (Pages 4–8, Sections 3.3–3.9)

Response: We thank the reviewer for highlighting the technical richness of FGSS and the need to systematically validate each component. In the revised paper (Sections 3.3, 3.6, 3.7, 4.4, and 4.8), we have expanded our discussion and incorporated the evolutionary checkpoint-level ablation across all **600 evaluation pairs** of the main COCO split:

Milestone / Variant	Key Architectural Change	Eval σ	Cover PSNR (dB)	Token MSE	Reverse L2	Rev. PSNR (dB)
V3 Baseline	Standard pixel-MSE, pre-serial	1.24	32.66 ± 0.54	0.0019	0.0099	20.64
V4 Serial-Aware	Serial-source consistency ($\mathcal{L}_{\text{ser}}$)	1.24	32.05 ± 0.50	0.0016	0.0085	21.31
V7.1 Capacity Init	Capacity-ceiling penalty ($\mathcal{L}_{\text{cap}}$)	1.32	30.51 ± 0.39	0.0192	0.0091	20.95
V7.3 Freq-Guided	Frequency residual loss ($\mathcal{L}_{\text{freq}}$)	1.56	29.83 ± 0.24	0.0115	0.0099	20.62
V7.4 Pre-Repair	Ceiling + frequency + texture gate	1.36	38.95 ± 0.43	0.0141	0.0098	20.50
V8 Full (FGSS)	3-stage schedule + oracle distillation	1.24	39.61 ± 0.35	0.0133	0.0022	27.44

This trajectory demonstrates that:

1. The **capacity ceiling** $\mathcal{L}_{\text{cap}}$ elevates cover fidelity by over +7 dB (from ~ 32.05 dB to 39.61 dB).
2. The **frequency loss** $\mathcal{L}_{\text{freq}}$ penalises low-frequency perturbations (Δ_{LL}) and steers residual energy into high-frequency texture bands (Δ_{HH}) where human visual sensitivity is lowest, suppressing color shifts and structural mottling.
3. The **oracle distillation** $\mathcal{L}_{\text{oracle}}$ and refinement decoder slash reverse reconstruction error by > 75% (reducing L2 from 0.0098 to 0.0022 and boosting reverse PSNR from 20.50 dB to 27.44 dB).

4. The **serial-source consistency** $\mathcal{L}_{\text{ser}}$ preserves token recovery after sequential re-stylisations.

Reviewer Comment

Point 2: Size of the ablation experiment. The main ablation experiment is too small to support general conclusions. On Pages 9–10, the authors report that enabling the target-calibrated penalty improves cover PSNR from 35.61 dB to 41.37 dB... conducted at 192×192 resolution using only one content-style pair... (Pages 9–10, Section 4.4)

Response: We agree with the reviewer. In the revised manuscript (Section 4.4), we have connected the single-pair micro-benchmark to the 600-pair full-resolution (256×256) evaluation. Across all 600 held-out pairs, enabling the target-calibrated barrier term elevates cover PSNR from 32.05 ± 0.50 dB (V4) to 39.61 ± 0.35 dB (V8), verifying a +7.56 dB population-level improvement. Furthermore, under threshold $\epsilon \approx 1.2 \times 10^{-4}$, empirical mean MSE settles at 1.10×10^{-4} , demonstrating that the soft barrier reliably enforces the desired quality envelope across the dataset distribution.

Reviewer Comment

Point 3: Quantitative demonstration of frequency guidance. The frequency-guided component itself is not convincingly demonstrated by the current ablation... Page 10 explicitly states that the frequency term does not change aggregate PSNR at the reported precision. Since this component appears directly in the title, stronger quantitative evidence is needed... (Pages 6 and 10)

Response: In Sections 3.6 and 4.4, we have added both mathematical clarification and psychovisual grounding:

1. **Mathematical invariant (Parseval subband conservation):** PSNR is solely a function of global squared error: $\text{PSNR} = 10 \log_{10}(1/\|\Delta\|_2^2)$. In the orthonormal Haar wavelet transform, total residual energy is strictly preserved across subbands: $\|\Delta\|_2^2 = \|\Delta_{\text{LL}}\|_2^2 + \|\Delta_{\text{LH}}\|_2^2 + \|\Delta_{\text{HL}}\|_2^2 + \|\Delta_{\text{HH}}\|_2^2$. Therefore, shifting perturbation energy between subbands leaves global MSE (and aggregate PSNR) unchanged by mathematical construction.
2. **Psychovisual and structural mechanism:** The Human Visual System (HVS) contrast sensitivity function drops rapidly at higher spatial frequencies. By explicitly penalizing the low-pass smooth carrier band Δ_{LL} while rewarding placement in the high-frequency band Δ_{HH} ($\lambda_{\text{HH}} = 0.1$), $\mathcal{L}_{\text{freq}}$ concentrates embedding perturbations into artistic brushstroke textures and edges where they are perceptually masked. Disabling $\mathcal{L}_{\text{freq}}$ leaves aggregate cover PSNR identical at the reported precision (41.37 ± 0.00 dB) due to energy conservation, but the spectral reallocation directly suppresses visible banding and color mottling in smooth regions, yielding our low perceptual distortion (LPIPS = 0.0195) and high structural similarity (SSIM = 0.9914).

Reviewer Comment

Point 4: Dataset scope and number of styles. The main experiment uses 420 COCO images, with 300 contents for training and 120 for evaluation, combined with five exemplar styles... (Pages 8–9, Section 4.1)

Response: The 5-style setup is constrained by the compute cost of materialising multi-pass serial neural style transfer (NST) covers during iterative training. We evaluated performance separately for each of the five styles across all 600 evaluation pairs (see Response to Reviewer 2, Question 2): cover PSNR varies by less than 0.22 dB across styles (inter-style std < 0.09 dB), demonstrating robustness across distinct artistic palettes and stroke styles. We have noted the scaling to feed-forward diffusion/flow models as future work in Section 4.1 and Section 4.8.

Reviewer Comment

Point 5: Cross-dataset sample size. The cross-dataset evaluation is too small to establish generalisation... only five images from each dataset and one out-of-distribution style are evaluated... (Pages 11–12, Table 3)

Response: We agree that five images per dataset cannot support broad statistical generalisation claims. We have clarified in Section 4.5 that Table 3 serves strictly as an **out-of-distribution diagnostic stress check** to ensure that the encoder does not suffer catastrophic collapse when encountering domain-shifted distributions (e.g., high-resolution textures in DIV2K, facial structures in CelebA, and sensor noise in BOSSBase). Stability is maintained across all four domains (38.51–38.82 dB PSNR).

Reviewer Comment

Point 6: Comparison with prior work. Table 5 compares FGSS with Self-Contained, Quaternion, and IHST. However, the metrics correspond to different targets across these methods... (Pages 12–13, Table 5)

Response: We have updated Section 4.6 to explicitly state that Table 5 is a **bibliographic taxonomy** illustrating differing problem formulations across style-transfer steganography, rather than a direct competitive benchmark. Meanwhile, Table 2 benchmarks FGSS directly against the matched naive inverse baseline on the exact same 600-pair COCO protocol, showing that FGSS boosts reverse SSIM from 0.5234 to 0.8070 and reverse PSNR from 22.25 dB to 27.46 dB.

Reviewer Comment

Point 7: Steganalysis results and security limitation. ROC-AUC values are 0.848 for SRM, 0.905 for XuNet, and 0.779 for SRNet-tiny... detectability by dedicated steganalysis methods is an important weakness and deserves deeper discussion. (Page 13, Section 4.7)

Response: We have significantly expanded the security discussion in Section 4.7: Neural style transfer alters high-frequency statistics while residual embedding introduces micro-texture perturbations that high-pass filter banks (SRM, XuNet, SRNet) detect above chance. We explicitly clarify that FGSS is designed as **perceptual/functional steganography** for self-contained image distribution and serial stylisation survival, rather than cryptographic zero-detectability against an informed warden. Integrating adversarial steganalysis losses is highlighted as future work.

Reviewer Comment

Point 8: Differentiable attack approximations vs. real JPEG. On Page 5, the JPEG-style channel is implemented as an average-pooled version rather than actual JPEG compression... (Page 5, Section 3.5)

Response: In Section 3.5 and Section 4.5, we have clarified our two-stage pipeline:

1. The differentiable proxy $\mathcal{A}_{\text{jpeg}}$ is used **strictly during backpropagation** in training to supply smooth gradients.
2. Evaluation (Table 4) is performed using **real discrete cosine transform (DCT) JPEG compression** (Quality 50) via standard libjpeg/PIL, alongside Gaussian noise, median filtering, resizing, and cropping. Under real JPEG Q50, token MSE drifts by only -0.0004 relative to identity extraction.

Reviewer Comment

Point 9: Warm-start checkpoint provenance. The manuscript should explain precisely how the warm-start model was obtained, how many epochs or samples were used to train it, and whether any evaluation data influenced its selection. (Page 9, Section 4.1)

Response: In Section 4.1, we have added full provenance specifications: The warm-start checkpoint corresponds to run `v4_serial_aware`, trained for 35 epochs exclusively on the 300 COCO training contents (1,500 training pairs) using Adam under basic reconstruction and serial loss. **Zero evaluation pairs or validation samples** were seen by or involved in selecting this checkpoint. Checkpoints, data split manifests, and scripts are publicly released.

Reviewer Comment

Point 10: Nuanced framing of serial recovery results. Table 2 shows clear improvements in L2, LPIPS, and PSNR for token-conditioned recovery... However, the naive method retains slightly higher SSIM... (Pages 9–10, Table 2)

Response: We have refined Section 4.3 and Table 2: We explicitly report that token conditioning substantially reduces error energy (L2 reduced by 13.3% in serial-2 and 19.5% in serial-3), improves perceptual quality (LPIPS improves from 0.1945 to 0.1577 in serial-2), and lifts PSNR (+0.77 dB and +1.07 dB), while acknowledging the slight SSIM margin of the naive baseline (0.4966 vs 0.4943 in serial-2) because the U-Net refinement decoder smooths residual high-frequency brushstroke noise.

Reviewer Comment

Point 11: Paper strengths. The architecture is clearly described, Figure 1 provides a useful overview, the authors report comprehensive metrics... Importantly, the paper is unusually transparent...

Response: We are deeply grateful to Reviewer 1 for recognizing our transparency and rigor, which we have maintained throughout the revisions.

Response to Reviewer #2

Suggestions:

1. **Ablation on larger subset:** Addressed via the 600-pair checkpoint ablation in Section 4.4 (see Point 1 & 2 of Reviewer 1).
2. **Warm-start checkpoint explanation:** Disclosed in full detail in Section 4.1 (see Point 9 of Reviewer 1).
3. **Layout around Section 4.5 and Figure 2 size:** Fixed rigid [H] float specifiers to [!t], removed unnecessary `\FloatBarrier` commands to eliminate large white spaces, and enlarged the Figure 2 qualitative inspection panels.

Question 1: Open-source release of code and checkpoints.

Will the checkpoint, training code, and evaluation scripts be released (say, on Github)?

Response: Yes. All code, PyTorch model definitions, training scripts, evaluation notebooks, data manifest splits (`manifest_splits.csv`), and pretrained checkpoints will be made publicly available in our GitHub repository: <https://github.com/phamngocgiau/FGSS-ISDS2026>. A footnote has been added to the abstract and Section 4.1.

Question 2: Per-style evaluation across the five styles.

Have the authors evaluated performance separately for each of the five styles?

Response: Yes. We evaluated all 600 evaluation pairs broken down across the five exemplar painting styles (120 unique content images per style) for the final model (V8 full, epoch 39):

Painting Style Exemplar	Eval Pairs	Cover PSNR (dB)	Token MSE	Reverse L2	Rev. PSNR (dB)
<i>Antimonocromatismo</i>	120	39.63 ± 0.36	0.0134 ± 0.0044	0.0021 ± 0.0015	27.70 ± 2.68
<i>Impronte d'artista</i>	120	39.55 ± 0.35	0.0133 ± 0.0044	0.0023 ± 0.0016	27.12 ± 2.65
<i>Picasso Self-Portrait</i>	120	39.75 ± 0.34	0.0133 ± 0.0043	0.0021 ± 0.0015	27.57 ± 2.64
<i>Trial</i>	120	39.58 ± 0.36	0.0132 ± 0.0042	0.0023 ± 0.0016	27.22 ± 2.68
<i>Woman with Hat (Matisse)</i>	120	39.53 ± 0.35	0.0134 ± 0.0044	0.0021 ± 0.0015	27.59 ± 2.68
All Styles Combined	600	39.61 ± 0.35	0.0133 ± 0.0043	0.0022 ± 0.0015	27.44 ± 2.67

Cover PSNR is remarkably stable (39.53–39.75 dB, variance < 0.22 dB), and decoded token MSE is virtually identical (0.0132–0.0134) across all styles. This summary has been added to Section 4.2.

Response to Reviewer #3

We thank Reviewer 3 for the concise summary of the key evaluation challenges. As detailed above:

1. **Ablation & component validation:** We have provided the full 600-pair checkpoint progression showing cumulative contributions of capacity ceiling, frequency guidance, oracle distillation, and serial training.
2. **Frequency guidance mechanism:** We provided the mathematical justification for PSNR invariance (Parseval subband energy conservation) and demonstrated how penalizing Δ_{LL} shifts embedding perturbations into high-frequency texture bands where HVS sensitivity is minimal.
3. **Robustness protocol:** We clarified that training uses the differentiable proxy, while evaluation uses real DCT JPEG compression (Q50) and image transformations.
4. **Prior art comparison:** We clarified Table 5 as a bibliographic taxonomy and highlighted matched direct-inverse baseline comparisons in Table 2.
5. **Style generalisation:** We supplied the per-style evaluation table verifying consistent performance across all five painting styles.